

4	(a) the energy (stored) in a body due to its extension/compression/deformation/ change in shape/size	B1 [1]
	(b) (i) two values of F/x are calculated which are the same e.g. $10.4/40 = 0.26$ and $6.5/25 = 0.26$	B1
	<i>or</i>	
	ratio of two forces and the ratio of the corresponding two extensions are calculated which are the same e.g. $5.2/10.4 = 0.5$ and $20/40 = 0.5$	(B1)
	<i>or</i>	
	gradient of graph line calculated and coordinates of one point on the line used with straight line equation $y = mx + c$ to show $c = 0$	(B1)
	(so) force is proportional to extension (and so Hooke's law obeyed)	B1 [2]
	(b) (ii) 1. $k = F/x$ or $k = \text{gradient}$	C1
	gradient or values from a single point used e.g. $k = 10.4/(40 \times 10^{-2})$	
	$k = 26 \text{ N m}^{-1}$	A1 [2]
	2. work done = area under graph or $\frac{1}{2}Fx$ or $\frac{1}{2}(F_2 + F_1)(x_2 - x_1)$ or $\frac{1}{2}kx^2$ or $\frac{1}{2}k(x_2^2 - x_1^2)$	C1
	= $\frac{1}{2} \times 10.4 \times 0.4 - \frac{1}{2} \times 5.2 \times 0.2$ or $\frac{1}{2} \times (5.2 + 10.4) \times 20 \times 10^{-2}$ or $\frac{1}{2} \times 26 \times (0.4^2 - 0.2^2)$	C1
	= 1.6 J	A1 [3]
	(c) remove the force and the spring goes back to its original length	B1 [1]